CCNA PRACTICE LABS

7 Before-and-After Comparisons

A visual record of the newly available dual-stack lab and six added CCNA practice labs. Each comparison shows the starting state beside the verified configuration.

7	6	1
labs documented	new lab scenarios	previously hidden lab unlocked

The existing four public labs are unchanged. Lab 5 is the previously hidden IPv4/IPv6 exercise; Labs 6-11 cover interface addressing, VLAN, LACP, ACL, routing, and OSPF topics.

Starting configurations and verified outcomes are presented for practice reference.

Lab 5 | IPv4 and IPv6 Router Connectivity

BEFORE | ORIGINAL STARTING STATE

STARTING STATE

```
R1# show ip interface brief
Ethernet0/1  unassigned  administratively down
R2# show ip interface brief
Ethernet0/1  unassigned  administratively down

IPv4 and IPv6 addresses are missing.
The point-to-point link is down; neither ping can succeed.
```

AFTER | VERIFIED CONFIGURATION

```
[R1]
interface e0/1
 ip address 192.168.180.17 255.255.255.240
 ipv6 address 2001:db8:acca::1/64
 no shutdown

[R2]
interface e0/1
 ip address 192.168.180.30 255.255.255.240
 ipv6 address 2001:db8:acca::2/64
 no shutdown

[VERIFY FROM R1]
ping 192.168.180.30  =>  100% success
ping ipv6 2001:db8:acca::2  =>  100% success
Both router configs saved.
```

VERIFIED IPv4 and IPv6 point-to-point connectivity verified.

Lab 6 | IPv4 /26 and IPv6 /125 Point-to-Point

BEFORE | ORIGINAL STARTING STATE

STARTING STATE

```
R1# show ip interface brief
Ethernet0/0  unassigned  administratively down
R2# show ip interface brief
Ethernet0/0  unassigned  administratively down

The /26 IPv4 and /125 IPv6 addresses are missing.
The routed link is down on both ends.
```

AFTER | VERIFIED CONFIGURATION

```
[R1]
interface e0/0
 ip address 10.26.0.1 255.255.255.192
 ipv6 address 2001:db8:26::1/125
 no shutdown

[R2]
interface e0/0
 ip address 10.26.0.2 255.255.255.192
 ipv6 address 2001:db8:26::2/125
 no shutdown

[VERIFY FROM R1]
ping 10.26.0.2  =>  100% success
ping ipv6 2001:db8:26::2  =>  100% success
Both router configs saved.
```

VERIFIED The /26 and /125 interfaces are up and reachable.

Lab 7 | VLANs and Access Port Membership

BEFORE | ORIGINAL STARTING STATE

STARTING STATE

```
SW-1# show vlan brief
No Staff or Guest VLANs exist.
SW-1 E0/1 and SW-2 E0/1 are not assigned.
The inter-switch link has no trunk VLAN policy.

PC1: VLAN 20 (Staff)
PC2: VLAN 30 (Guest)
```

AFTER | VERIFIED CONFIGURATION

```
[BOTH SWITCHES]
vlan 20
 name Staff
vlan 30
 name Guest

[SW-1 E0/1]
switchport mode access
switchport access vlan 20

[SW-2 E0/1]
switchport mode access
switchport access vlan 30

[BOTH E0/0 TRUNK PORTS]
switchport mode trunk
switchport trunk allowed vlan 20,30
Switch configs saved.
```

VERIFIED Both VLANs, access ports, and the trunk are configured.

Lab 8 | Access Ports and LACP Trunk EtherChannel

BEFORE | ORIGINAL STARTING STATE

STARTING STATE

```
SW-1 and SW-2 ports are in dynamic mode.
E0/0-E0/1 are not bundled; no port-channel is active.
E0/2 endpoint ports are not assigned to VLAN 40.

Expected bundle: Po10
SW-1: LACP active
SW-2: LACP passive
```

AFTER | VERIFIED CONFIGURATION

```
[BOTH E0/2 ACCESS PORTS]
switchport mode access
switchport access vlan 40

[BOTH E0/0-E0/1 MEMBER TRUNKS]
switchport mode trunk
switchport trunk allowed vlan 40,50

[SW-1]
interface range e0/0 - 1
channel-group 10 mode active

[SW-2]
interface range e0/0 - 1
channel-group 10 mode passive
Po10 members and configs verified.
```

VERIFIED Access VLAN and active/passive LACP are configured.

Lab 9 | Standard ACL: Permit a Trusted Host

BEFORE | ORIGINAL STARTING STATE

STARTING STATE

```
R1 has no ACL on its LAN interface.
R1 and R2 have no routes between the LANs.

Trusted PC  192.168.10.10
Guest PC    192.168.10.20
Server      192.168.50.10

Neither source is filtered.
```

AFTER | VERIFIED CONFIGURATION

```
[R1 ROUTING + ACL]
ip route 192.168.50.0 255.255.255.0 10.0.12.2
access-list 10 permit host 192.168.10.10
interface e0/0
 ip access-group 10 in

[R2 RETURN ROUTE]
ip route 192.168.10.0 255.255.255.0 10.0.12.1

[VERIFY]
Trusted PC -> server: PERMITTED
Guest PC -> server: BLOCKED (implicit deny)
Both router configs saved.
```

VERIFIED Trusted host reaches the server; guest traffic is filtered.

Lab 10 | Static and Default Route Forwarding

BEFORE | ORIGINAL STARTING STATE

STARTING STATE

```
R1 and R2 know only their connected networks.
R1 has no route to 172.16.20.0/24.
R2 has no route back to 172.16.10.0/24.

PC1  172.16.10.10/24
R1 -- R2  10.10.12.0/30
Server  172.16.20.20/24
End-to-end ping fails.
```

AFTER | VERIFIED CONFIGURATION

```
[R1 SPECIFIC ROUTE]
ip route 172.16.20.0 255.255.255.0 10.10.12.2

[R2 DEFAULT ROUTE]
ip route 0.0.0.0 0.0.0.0 10.10.12.1

[VERIFY]
show ip route
ping 172.16.20.20  =>  100% success
Return traffic uses R2's default route.
Both router configs saved.
```

VERIFIED Specific and default routes provide end-to-end connectivity.

Lab 11 | OSPF DR Election on a Broadcast LAN

BEFORE | ORIGINAL STARTING STATE

STARTING STATE

```
R1, R2, and R3 share 10.11.0.0/29.
All interfaces are up; no OSPF process is configured.
No router IDs or interface priorities are set.

R1 default priority: 1
R2 default priority: 1
R3 default priority: 1
No DR election can occur yet.
```

AFTER | VERIFIED CONFIGURATION

```
[ALL ROUTERS]
router ospf 10
 network 10.11.0.0 0.0.0.7 area 0

R1 router-id 1.1.1.1; priority 100
R2 router-id 2.2.2.2; priority 150
R3 router-id 3.3.3.3; priority 150

[INITIAL ELECTION]
R3 wins: priority 150 ties R2, then router ID
3.3.3.3 wins over 2.2.2.2.
All router configs saved.
```

VERIFIED R3 wins the initial DR election by router-ID tie-break.